OPERATING SYSTEMS (CS304)

Complete Syllabus

Course Description

This course introduces the concepts, principles, and components of modern operating systems. Students learn process management, CPU scheduling, deadlocks, memory management, file systems, and disk scheduling.

Course Objectives

• Understand the role and functions of an Operating System.
• Learn process and thread management.
• Analyze CPU scheduling algorithms.
• Understand deadlocks and prevention techniques.
• Study memory management and virtual memory.
• Explore file systems and storage management.

Unit 1 - Introduction & Processes

Introduction to Operating Systems, Functions of OS, Process Concept, Process States, Process Control Block, Threads, Multithreading, and System Calls.

Unit 2 - CPU Scheduling

Scheduling Concepts, FCFS Scheduling, SJF Scheduling, Round Robin Scheduling, Priority Scheduling, Scheduling Criteria, and Synchronization Basics.

Unit 3 - Deadlocks

Deadlock Characterization, Necessary Conditions, Resource Allocation Graph, Deadlock Prevention, Avoidance, Detection, Recovery, and Banker's Algorithm.

Unit 4 - Memory Management

Memory Allocation, Paging, Segmentation, Virtual Memory, Demand Paging, Page Replacement Algorithms, and Memory Protection.

Unit 5 - Storage & File Systems

File Concepts, File Allocation Methods, Directory Structures, Disk Structure, Disk Scheduling Algorithms, and Secondary Storage Management.

Course Outcomes

CO1: Explain operating system concepts and services.
CO2: Analyze process and thread management techniques.
CO3: Apply CPU scheduling and synchronization concepts.
CO4: Evaluate memory management techniques.
CO5: Understand file systems and storage management.

Assessment Pattern

Internal Assessment, Assignments, Quizzes, Lab Work, and End Semester Examination.

Recommended Textbooks

1. Operating System Concepts – Silberschatz, Galvin.
2. Modern Operating Systems – Andrew Tanenbaum.

Summary

The Operating Systems course provides foundational knowledge required to understand how computer systems manage resources and execute applications efficiently.